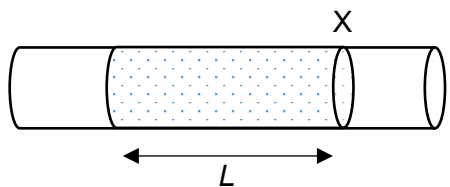
19 A copper wire has a number density of 8.5×10^{28} conduction electrons per cubic metre, and an cross-sectional area of 3.2×10^{-7} m^2. When a potential difference is applied to the ends of the wire, the current is 1.0 A.



If all the electrons within a cylinder of length *L* pass point X in 60 s, what is the value of *L*?

A 0.00015 m **B** 0.014 m **C** 0.025 m **D** 0.20 m